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WATER ARCHITECTURE IN SOUTH ASIA

A Study of Types, Development and Meanings

BY

JULIA A.B. HEGEWALD

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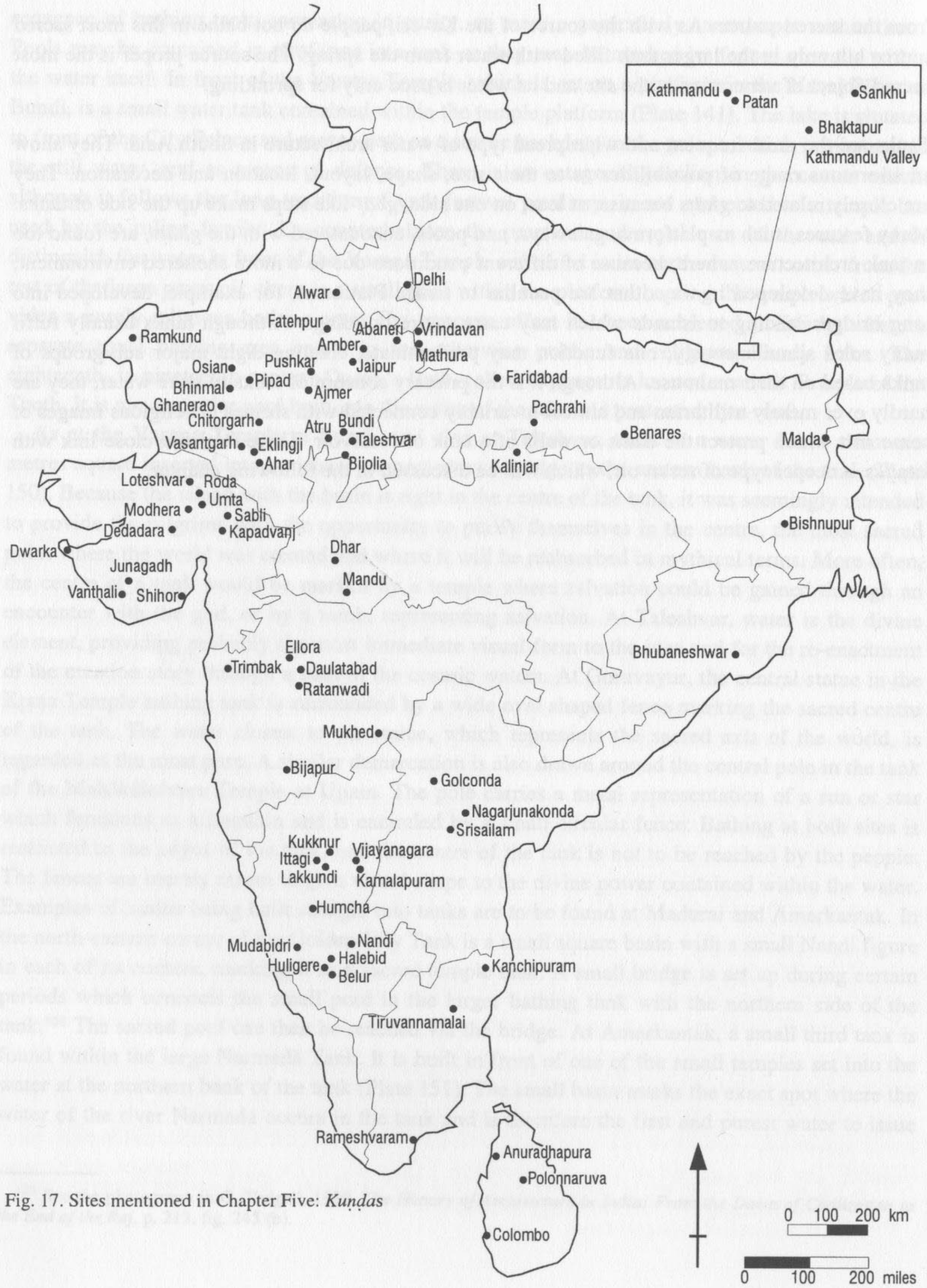

Fig. 17. Sites mentioned in Chapter Five: *Kuṇḍas*

CHAPTER FIVE

KUNDAS

1. INTRODUCTION

Kuṇḍas[1] are stepped water basins which penetrate deep into the ground and in section resemble the shape of a funnel with a relatively small water area at the bottom (Fig. 18). Most *kuṇḍas* are fed by ground water, shallow aquifers or springs. The sides of *kuṇḍas* are usually set at a steep angle, often greater than the natural slope of the ground.[2] As a consequence, they need a large mass of steps, often in complicated pyramidal or triangular sets, to buttress the terraced walls. If they were of insufficient strength to counteract the inward thrust of the side walls, the stepped sides of the *kuṇḍas* would be likely to slip away.

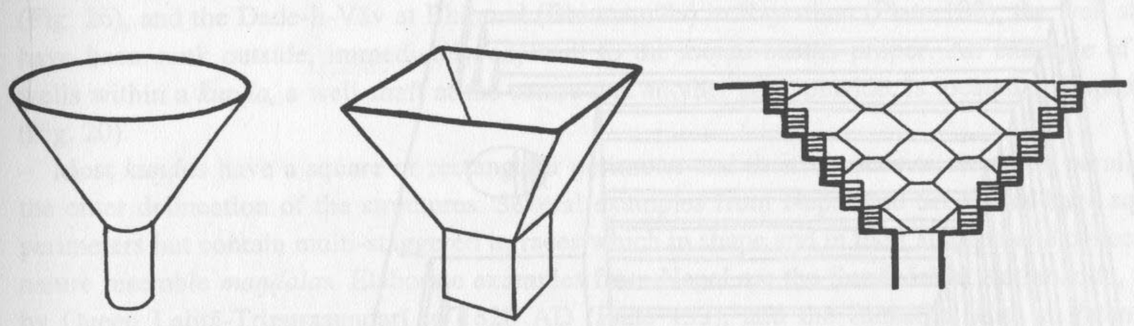

Fig. 18. *Kuṇḍas* have deep funnel-shaped basins

Steep vertical sides are more easily constructed in hard soil or rock than in sandy ground, and therefore soil conditions have a strong influence on the shape of *kuṇḍas*. The more precipitous the sides of a *kuṇḍa*, the larger the water surface can be at the bottom of the pit. The water area also depends on the depth of the *kuṇḍa* and its top perimeter. Because the water level in the funnel fluctuates, the water surface area is variable. Owing to the considerable depth and relatively small

[1] *Kuṇḍa* is the Sanskrit version of the Hindi term *kuṇḍ*. The general type of water architecture has been given the traditional Sanskrit term *kuṇḍa*, and for reasons of consistency all named *kuṇḍas* in this study have been referred to using the Sanskrit version. In colloquial language, many people in South Asia will refer to these structures by using the Hindi version, leaving out the final '-a', whilst priests and texts are more likely to retain the final '-a' when discussing such structures. In using the Sanskrit version of the term, it is not intended to impose a sanskritic stamp on structures found in the south of India. This study aims at showing parallels and commonalties between water structures in South Asia. Therefore, it did not seem to make sense to differentiate between various regional versions of the term, where the structures are clearly of one type, and the Sanskrit version was perceived to be the most neutral.

[2] This is for instance the case in the *kuṇḍa* at Kapadvanj, where the angle of the sides is about 38° to the horizontal, while the natural slope of the soil is circa 32°. (P. K. Patel, 1973, "Structural and Constructional Aspects of Water-Reservoirs and their Relationship with Religious Buildings in Gujarat", diploma thesis, p. 79).

water area, surface evaporation from *kuṇḍas* is low in comparison to tanks. But the steep angle of the funnel-shaped sides and their multi-stepped nature are the most characteristic elements of the *kuṇḍas,* distinguishing them clearly from tanks whose gently sloping short sides frequently lead to large water basins.[3] *Kuṇḍas* have such a striking aesthetic that their shape and step formations have often been 'imitated' in structures which actually function as tanks by collecting rain water or being fed by channels (Plate 162). Compared to *kuṇḍas* which draw ground water, these examples do not penetrate as deep into the ground and often have larger water basins at the bottom. They may, however, be classified as *kuṇḍas* because, when they are filled with water, obscuring the solid floor or the well shaft in the middle, their technical function cannot be detected. The key factor is that the builders intended these structures to look like *kuṇḍas.* As this study is primarily an art-historical enquiry into the choice and the development of shapes and the continuity of themes and visual forms, and not a detailed analysis of the functional engineering characteristics of water architecture, visual appearance has been the determining factor in the classification of these water structures as *kuṇḍas.*

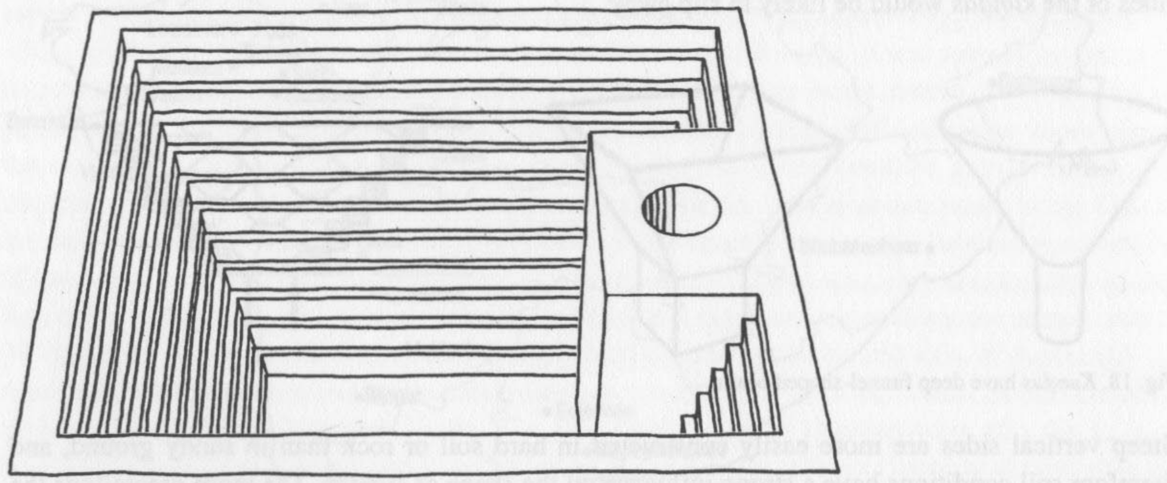

Fig. 19. *Kuṇḍa* in which the well shaft has been integrated into one of the sloping sides

Kuṇḍas which draw ground water usually have a well shaft in the centre of their basins. They are clearly distinguished from wells proper by the fact that in a *kuṇḍa,* the central well opens up into a funnel in the shape of an inverted pyramid at surface level, usually lined by precipitous steps. This funnel-shaped upper section is not a feature of ordinary wells and stepwells, and consequently is one of the most essential visual elements in the definition of a *kuṇḍa.* Some examples which have central square wells are the large *kuṇḍa* in Ghanerao near Ranakpur (Plate 152), the Amṛt Sarovar in the hill fort of Nandi north of Bangalore (Plate 159), the *kuṇḍa* outside the wall of the Mallikārjuna Temple complex at Srisailam, and the *kuṇḍa* of the Rāmeśvara Temple[4] at Bhubaneshwar. Because the ground water table at the latter two sites appears to have dropped considerably, and

[3] Tanks have been discussed in detail in Chapter Four.
[4] This temple is also known as Mausimaha Temple.

the water rarely reaches the funnel-shaped stepped sides of the *kuṇḍas,* modern water pumps have been installed using the old well shafts in the centres to reach the reduced ground water level. In other examples, small wells of only one or two metres in diameter, can be found in one corner of a *kuṇḍa*-basin. Here, the funnel shape of the stepped sides is not open at the bottom, but has a solid floor, with a small opening through which ground water is drawn. The Sūrya Kuṇḍa at Modhera, the Maṅgala Tīrtha[5] of the Ekāmbareśvara Temple at Kanchipuram (Plate 154), and the Kva Hiṭī at Kathmandu, follow this layout.

In other cases, the well shaft has been sunk into one of the sloping sides of the *kuṇḍa* or positioned adjacent to it; the deep basins serve as water storage units. From the draw well next to the basin, the ground water is pulled up and used from buckets; the *kuṇḍa*-basin is used for bathing and washing, the people descending to the water by means of the characteristic steps. By having a well and a basin, clean fresh drinking water in the well is separated from water for washing and bathing. It is usual for these side well shafts of *kuṇḍas* to be covered at the top to keep them clean and cool, whilst access to the well is from the side, within the *kuṇḍa* below ground level. Whereas at the Rajasthani sites of Osian (Osiāñ) and Abaneri, the well shafts have been built into the stepped east and north sides of the *kuṇḍa* respectively (Fig. 19), at the Lolārka Kuṇḍa at Benares (Fig. 26), and the Dade-li-Vāv at Bhinmal (Bhinnamāla) in Rajasthan (Plate 188), the well shafts have been sunk outside, immediately adjacent to the *kuṇḍa*-basins proper. An example of two wells within a *kuṇḍa,* a well shaft at the centre and another sunk outside, is located at Kapadvanj (Fig. 20).

Most *kuṇḍas* have a square or rectangular perimeter and their terraces or steps run parallel to the outer delineation of the structures. Several examples from Nepal and Sri Lanka have square perimeters but contain multi-staggered terraces which in shape and in their stepped and concentric nature resemble *maṇḍalas.* Elaborate examples from Nepal are the Sundhārā at Kathmandu, built by Queen Lalitā-Tripurasundarī in 1828 AD (Plate 153), and the Jaulākhel Hiṭī at Patan.[6] In contrast, much simpler *kuṇḍas* can also convey a strong diagrammatic character through their rigid symmetry and repetition of forms in diminishing planes. The clearest example of such a *maṇḍala*-shaped basin from Sri Lanka is the Kumāra Pokuna at Polonnaruva (Plate 190). It is interesting to observe the close relationship between water structures in Nepal and Sri Lanka. Strong similarities between the water architecture of the two countries are evident not only from their choice of shapes, but also in the material, which is usually burnt brick integrating stone steps and cornices, and in the elaborate water spouts often in animal shapes.[7] Because all the main Sri Lankan water structures and many of the Nepali ones are Buddhist, a reason for the strong similarities may lie in the transmission of Buddhism, and in this context the close relationship between stupas in Nepal and Sri Lanka may also be mentioned. A reason for the fact that Sri

[5] The local Tamil version of the name of this *kuṇḍa* is Maṅgala Tīrtham.
[6] A further noteworthy aspect of this deep and large *kuṇḍa* is that it contains a platform carrying a stone image of Viṣṇu-Nārāyaṇa set into a small shallow basin which is raised on a high pedestal in the centre of the *kuṇḍa.* Similar shallow basins containing images of Viṣṇu have been discussed in connection with *ghāṭs* and tanks in the relevant chapters.
[7] For a detailed discussion of these structures, see the section on conduit-*kuṇḍas* later in this chapter.

Lankan water structures are so different from south Indian tanks may be that the Sri Lankan people aimed at differentiating themselves from the style of the Tamils, who over many centuries regularly invaded their country.

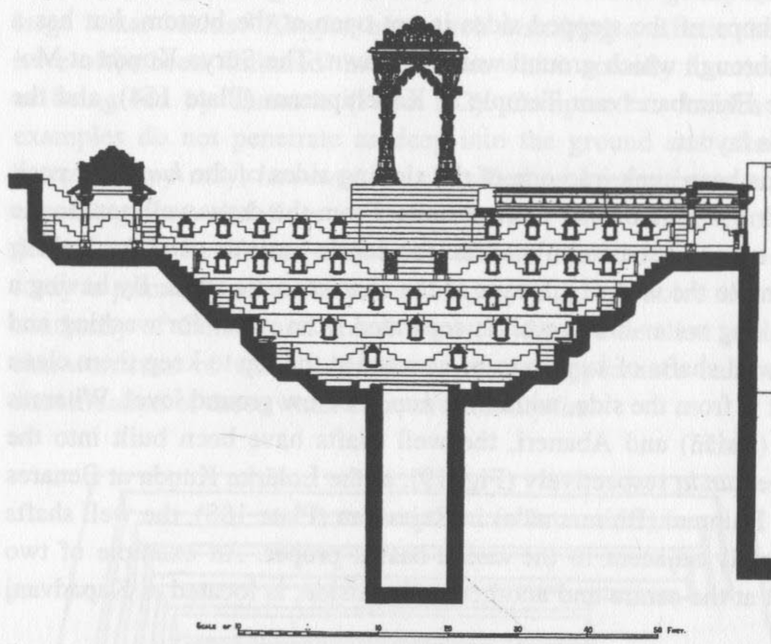

Fig. 20. Kapadvanj: the *kuṇḍa* has a well shaft in its centre and one sunk outside the basin (Burgess, 1905)

According to a Newari architectural manual even the underground pipes of the local *kuṇḍas* should be arranged in the shape of a cosmic diagram such as a svastika.[8] Other *kuṇḍas* have been constructed in lotus shapes, as may be seen in the Lotus Pool at Polonnaruva (Plate 46). Examples from Nepal are the Palikhyaḥ Hiṭī on the main road of Bhaktapur, the Jagati Gahiṭī in the Vajra Yoginī Sanctuary at Sankhu (Sāṃkhu), and the Bhagvati Temple Dhārā in Kathmandu. The small ablution *kuṇḍa* of the Aśvamedha Complex at Nagarjunakonda, which also resembles a lotus, is usually referred to as Kūrma Tank, meaning tortoise tank. By constructing *kuṇḍas* on *maṇḍala*, lotus or tortoise plans, which are all regarded as supports of the cosmos, the water structures were firmly related to the foundation of the universe. The Brahmā-jī-kā Kuṇḍa at Dwarka has a basin which is square at the bottom and goes over into a twelve-cornered shape at a higher level (Plate 155). The Sūraj Kuṇḍa south of Delhi is an unusual example of a large oval shaped *kuṇḍa*, whilst other examples are irregularly shaped to fit a given terrain, particularly if located in densely constructed hilltop forts, such as the Kukadeśvara Kuṇḍa in Chitorgarh.

It is a significant feature that, when regularly formed, almost all *kuṇḍas* are precisely oriented on the cardinal points of direction, tying them even closer into the structure of the universe. Very few *kuṇḍas* have even a slight error from exact orientation and even fewer have no orientation at all. But an interesting example of the latter are the twenty-seven sacred water basins, most of them *kuṇḍas*, along the fourteen kilometres long *pradakṣiṇā-patha* surrounding the sacred mountain at Tiruvannamalai. These are not oriented towards the cardinal directions, but towards the central mountain, the sacred centre of the site. The pilgrimage site is associated with Śiva in the shape of fire *liṅga*. The concept of indicating the four directions of space has also been frequently expressed in the fabric of the *kuṇḍas* by having four ramps, as in the Potarā Kuṇḍa at Mathura (Plate 156), or four flights of steps leading up from the water, as may be seen in the *kuṇḍa* of the Kāmākṣī Temple at Kanchipuram, and the Viśvanātha Kuṇḍa at Ellora.[9] In the Kāpaḍvañj Kuṇḍa, three pavilions and one *toraṇa* gateway indicate the orientation of the structure towards the cardinal points, and in the *kuṇḍa* at Ratanwadi (Ratanwāḍī) in Maharashtra, the central section of three sides is slightly recessed, while on the fourth side it has a central flight of steps leading up from the basin.

[8] R. O. A. Becker-Ritterspach, 1995, *Water Conduits in the Kathmandu Valley*, p. 25.

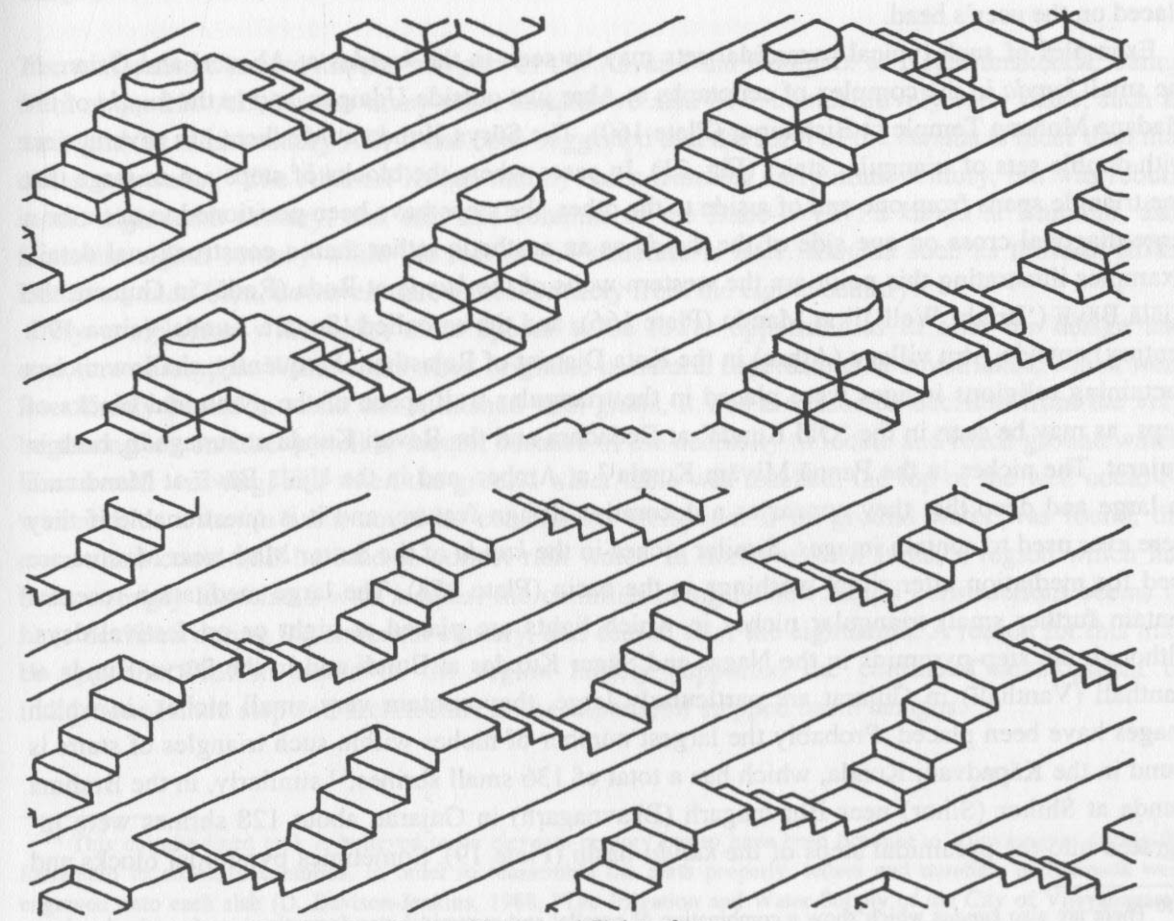

Fig. 21. Steps arranged in triangular blocks with bilateral flights of steps

[9] There are many more examples of *kuṇḍas* all over India, which have steps pointing into three of the cardinal points of direction and a protruding platform for lifting up water, marking the mid-point of the fourth side. One such example is the *kuṇḍa* outside Faridabad (Farīdābād) in Uttar Pradesh. *Kuṇḍas* with such projections will be discussed in detail in the section on *kuṇḍas* with risalits.

Steps in *kuṇḍas* may be long and parallel to each other, as can be seen in the small Gaurī Kuṇḍa on Kedār Ghāṭ at Benares, or in the Amṛt Sarovar in the fort at Nandi. The latter has larger tiers of wider terraces at regular intervals structuring the long descent to the water contained in the central well. Some *kuṇḍas* have small outcropping platforms which are more typical of large tanks. Examples are the Lolārka Kuṇḍa at Benares, the Undeśvara Kuṇḍa at Bijolia (Bijauliyā) in Rajasthan, and the small *kuṇḍa* at Maṇikarṇikā Ghāṭ at Benares (Plate 157). However, the most usual feature of *kuṇḍas* is to have steps arranged in triangular blocks with bilateral flights of steps, where the stepping is directed in a series of diagonals across the side (Fig. 21).[10] Through this arrangement, the sides can be made much stronger and therefore steeper. Other practical considerations are favoured through steps constructed in triangular sets: the pyramidal blocks of steps create resting platforms at regular intervals; they prevent a person from falling down the entire length of the stairs; they can also be used to rest water pots and baskets which can then easily be placed on the user's head.

Examples of such typical pyramidal sets may be seen in the *kuṇḍas* at Abaneri and Osian, in the small *kuṇḍa* in the complex of cenotaphs at Ahar just outside Udaipur, and in the *kuṇḍa* of the Madana Mohana Temple at Bishnupur (Plate 160). The Sūrya Kuṇḍa at Modhera has several tiers with double sets of triangular stairs (Fig. 21). In cases where the blocks of steps are so large that one triangle spans from one end of a side to the other, the steps have been positioned to describe a large diagonal cross on one side of the *kuṇḍa* as an aesthetic rather than a constructional detail. Examples illustrating this point are the western walls of the *kuṇḍa* at Roda (Roḍā) in Gujarat, the Ujālā Bāvlī ('Bright Well')[11] at Mandu (Plate 166), and the so-called 'Square Kuṇḍa' (circa 19th century) outside Atru village (Aṭhru) in the Kota District of Rajasthan. Frequently, shallow niches containing religious images were placed in the triangular wall space of the pyramidal blocks of steps, as may be seen in the 'Old Kuṇḍa' at Dedadara and the Revaṭi Kuṇḍa at Junagadh, both in Gujarat. The niches in the Pannā Miyāṃ Kuṇḍa[12] at Amber, and in the Ujālā Bāvlī at Mandu are so large and deep that they appear as a decorative design feature, and it is questionable if they were ever used to contain images. Similar niches in the *kuṇḍa* of the Suttur Maṭh near Mysore are used for mediation after ritual washings in the basin (Plate 158). The large meditation recesses contain further small triangular niches in which lights are placed at night or on festival days. Although the step-pyramids in the Nagar and Sāgar Kuṇḍas at Bundi and in the Sūrya Kuṇḍa at Vanthali (Vanthalī) in Gujarat are particularly large, they contain very small niches, in which images have been placed. Probably the largest number of niches within such triangles of stairs is found in the Kāpaḍvañj Kuṇḍa, which has a total of 136 small shrines;[13] similarly, in the Brahmā Kuṇḍa at Shihor (Sihor) near Bhavnagarh (Bhavnagaṛh) in Gujarat, about 128 shrines were integrated into the pyramidal steps of the *kuṇḍa*-basin (Plate 19). Sometimes pyramidal blocks and

[10] There are also *kuṇḍas* which show a combination of parallel and pyramidal step formations, such as the *kuṇḍa* at Ghanerao.

[11] This *kuṇḍa* is called 'bright' because is it open, contrasting with another example near by which is covered and therefore called Andherī Bāvlī, meaning 'dark well'. (D. R. Patil, 1992, *Mandu*, pp. 28–29). The spellings *bāolī* or *bāoṛī* are also encountered in association with these structures.

[12] At times also spelled Pannā Mia or Mian Kuṇḍa.

[13] J. A. S. Burgess, 1905, "Muhammadan Architecture of Ahmadabad", p. 94.

long parallel steps were combined within one structure, as may be seen in the Sūrya Kuṇḍa at Vanthali, in the Khatan Bāvlī in the fort of Chitorgarh and the so-called 'Ritual Bath', near the Mahānavami Platform at Vijayanagara (Plate 162).[14] The *kuṇḍa* of the Ibrahim Rauzā at Bijapur has only two sets of pyramidal stair blocks positioned one above the other, on the east and the west sides.[15] A similar arrangement is found on the north and east sides of the bathing *kuṇḍa* of the Trymbakeśvara Temple at Trimbak (Plate 161). Here, the sets of pyramidal steps seem to go over into each other creating a large fan-shaped design.[16] A further example of this feature may be seen in the large rectangular *kuṇḍa* in front of the Āthār (Asar) Mahal at Bijapur. *Kuṇḍas* whose walls are structured by large tiers of terraces which contain narrow flights of small steps can be seen in the Potarā Kuṇḍa at Mathura, the large *kuṇḍa* east of the Laṅka Tilaka Image House at Polonnaruva, and these are typical of most *kuṇḍas* at Tiruvannamalai, and in Nepal (Plates 185, 198).

The two earliest known *kuṇḍas* are part of the Aśvamedha Complex at Nagarjunakonda, dating from about 220 AD, but for many other *kuṇḍas* we also have comparatively early dates, such as the eighth or ninth century AD. It has been suggested that the form of the *kuṇḍas* is older than that of the stepwells.[17] The Ābānerī Kuṇḍa mainly dates from the early ninth century, but was rebuilt in the eighteenth century, and only the columns of the Dade-li-Vāv, a *kuṇḍa* at Bhinmal, date from the eighth century while the rest of the structure is later. *Kuṇḍas* such as those at Roda, Dedadara and Osian, however, date almost entirely from the eighth century.

By comparison with tanks, there appear to be fewer opportunities to alter the design and structure of *kuṇḍas*. While tanks often originate in natural depressions or small lakes, which were later firmly bound in stone and furnished with *ghāṭs*, it was a conscious decision from the very beginning to construct a well or *kuṇḍa*, because of the necessity to locate and reach ground water. First a well was dug, and when the ground water table was reached, the top of the well could be widened, or a *kuṇḍa*-basin could be constructed alongside. If no ground water was found, the excavation could still be used to collect rain water. In north-western India, a region which has been strongly associated with *kuṇḍas*, the commissioning of new *kuṇḍa* constructions seems to have declined during the sixteenth century, and ceased after the eighteenth. A reason for this may be that the Muslim rulers of the region largely supported the continued development of indigenous Hindu stepwell architecture at the expense of stepped basin designs.

[14] This chlorite-lined tank is believed to be eleventh-century but to have been brought to Vijayanagara during the fourteenth to sixteenth centuries. In order to reassemble the parts properly, letters and numbers in Kannada were engraved onto each slab (D. Davison-Jenkins, 1988, "The Irrigation and Water Supply of the City of Vijayanagara", PhD thesis, pp. 69–70).

[15] The maze pattern decorating the sides of the doors to the tomb are believed to depict a plan of the system of water channels of the city of Bijapur (S. Gole, 1989, *Indian Maps and Plans: from Earliest Times to the Advent of European Surveys*, pp. 22–23).

[16] They are similar to the fan-shaped steps along *ghāṭs* discussed in Chapter Three.

[17] M. Livingston, 1995, "The Stepwells and Stepped-Ponds of Western India", p. 5. Stepwells will be discussed in the next chapter.

Kuṇḍas are typically found in towns, villages and the domestic quarters of forts, where they serve as public water places and are used for washing, bathing, and watering animals and fields. They also form part of palace compounds, where they are usually of a more ornamental type, not drawing ground water but simply illustrating the aesthetic of this type of water structure.[18] *Kuṇḍas* are also commonly found in connection with religious institutions, such as Hindu temples and Buddhist monasteries, where their sacred waters are used for ritual ablutions. They may be found in association with Islamic tombs, as for example at the Quṭb Shāhī Tombs at Golconda, and the Ibrahim Rauzā, Bijapur, but also with cenotaphs, such as the *kuṇḍa* at Ahar (Plate 163).[19] A few examples of *kuṇḍas* are located in mosque courtyards, as in the Jāmi Masjid at Malda in Bihar. Miniature *praṇālī-kuṇḍas* are typical of Karnataka (Plate 164).[20]

Kuṇḍas are regarded as particularly sacred among water structures, and are therefore frequently connected with religious buildings, with shrine rooms and small temples often integrated into their sides. The term *kuṇḍa* can sometimes be used to denote a sacred water site in general, and in Sanskrit sources there is often no clear differentiation between the terms *kuṇḍa* and *tīrtha*.[21] A reason for this emphasis on the religious significance of *kuṇḍas* may derive from the fact that they penetrate deep into the ground and open it up in the shape of a funnel flooded with light. The ground is believed to be inhabited by demons and supernatural beings which may be upset by such disturbances, and the well shafts connected with *kuṇḍas* may lead to the underworld. Therefore, shrines and religious images were placed around *kuṇḍa*-basins to ward off evil powers emerging from the water. Alternatively, water is sacred and so are the structures connected with it, frequently turning water architecture into open-air shrines and important places of pilgrimage. A more practical consideration is that *kuṇḍas* are particularly found in arid regions and desert areas, where water is a rare and precious commodity, regarded by the people as a gift from the gods who in return have to be pleased and comforted to ensure a continuous flow of water.

2. VARIATIONS IN KUṆḌAS CONSTRUCTION AND DESIGN

Kuṇḍas exist in various shapes, sizes and degrees of elaboration. The simplest and most common type of *kuṇḍa* has steps along all four sides of its basin. Such regular symmetrical and concentric *kuṇḍas* can be very small, as for instance the circa seven metres square *kuṇḍa*[22] at the Nīlakaṇṭha Temple in Kalinjar Fort, the thirteen and a half by seventeen metres Maṇikarṇikā Kuṇḍa at Maṇikarṇikā Ghāṭ in Benares, and the approximately thirteen by sixteen metres *kuṇḍa* in the ruins in front of the Rāmacandra Temple at Vijayanagara. Examples from all over South Asia of much larger *kuṇḍas* are the seventeenth-century *kuṇḍa* at Mudabidri (Muḍabidri, Mūrabidrī) in Karnataka, the circa fourteenth to fifteenth-century Elephant Tank at Daulatabad, the fourth- to sixth-

[18] A good example is the so-called 'Ritual Bath' at Vijayanagara.
[19] Outside the tomb complex, there are two more *kuṇḍas* associated with *śaiva* worship.
[20] For a brief discussion of *praṇālī-kuṇḍas* and tanks see Chapter Four.
[21] A. W. Entwistle, 1987, *Braj: Centre of Krishna Pilgrimage*, p. 293.
[22] All measurements relate to the lengths of the top perimeter of the *kuṇḍas*.

century Ath Pokuna (Elephant Pond) at Anuradhapura, and the early eleventh-century Sūrya Kuṇḍa at Modhera (Plate 167), the latter measuring about 360 by 540 metres.

Kuṇḍas with Risalits

Other *kuṇḍas* have steps only on three sides while the fourth is a plain wall (Fig. 22 b). A very small and simple example of this version is the Gaurī Kuṇḍa on Kedār Ghāṭ at Benares (7.5 by 10 metres). Its eastern side is straight with a small niche in its centre which may be used as an outlet into the Ganges. A larger example is the old *kuṇḍa* at the site for the new Digambara Pārśvanātha Jaina Temple at Bijolia, measuring about twenty-two and a half metres square. While three sides

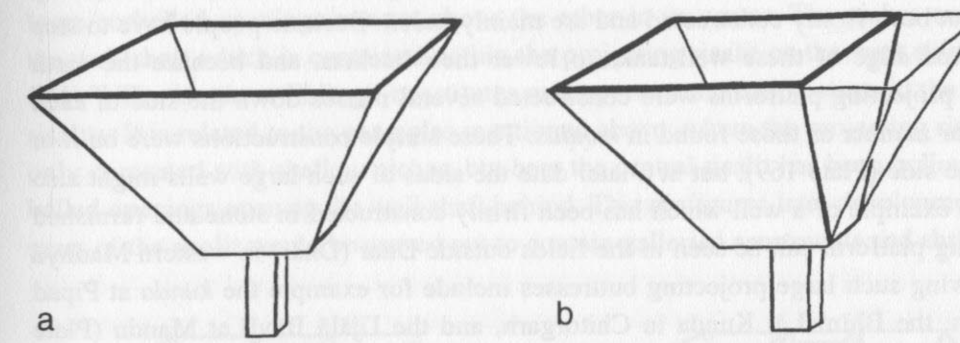

Fig. 22. *Kuṇḍa* with four sloping sides (a), and *kuṇḍa* with three sloping sides and one vertical wall (b)

have pyramidal step formations, the north side is plain with a small outlet channel in its centre. A large Digambara *mānastambha* was set next to the basin at its north-western corner. At Dedadara, one of the long sides of the rectangular 'Old Kuṇḍa' (circa eighth century AD)[23] is a plain vertical wall (Plate 165). This *kuṇḍa* is the place of an annual fair in which unmarried women fast at the sacred water site in the hope of finding a good husband. A similar architectural arrangement may have existed at Roda, where the long northern side of the rectangular *kuṇḍa*, dating from the eighth century AD, has entirely subsided (Plate 168). It is likely that this side wall too was a long and vertical wall,[24] which collapsed because it was not buttressed by a mass of steps as on the other three sides, which are still in good condition. Although the water basin of the Kacchapeśvara Svāmī Temple at Kanchipuram is not very deep, three sides were furnished with steps and the fourth side, to the west, is a vertical wall, typical of *kuṇḍa* constructions. In the centre of this wall is a square recess in which is a small plain shrine containing a *liṅga*. The space surrounding

[23] Livingston, 1995, p. 14. Although the dating by the AIIS, at Gurgaon, is slightly later (ninth or early tenth century AD), the earlier circa eighth century dating seems reasonable with regard to the dating of the similar eighth century Roḍā Kuṇḍa.
[24] M. A. Dhaky's plan of a reconstructed *kuṇḍa* at Roda depicts this northern wall with pyramidal steps mirroring the opposite site, but he does not give reasons for this assumption (M. W. Meister, & M. A. Dhaky, 1991, *Encyclopaedia of Indian Temple Architecture—North India: Period of Early Maturity (c. AD 700–900)*, p. 353, fig. 164).

the shrine can be used as a narrow *pradakṣiṇā-patha.* Such circumambulation paths also surround the larger shrines in the centre of the sides in the Sūrya Kuṇḍa at Modhera (Plate 23).

Often, one finds a pulley mechanism positioned on these straight walls, as at the Jaina temple at Bijolia and in the Roḍā Kuṇḍa, where a bucket is pulled up over a wheel. Through this feature, a *kuṇḍa* without an independent well shaft can be used for both purposes, pulling up buckets or descending to bathe in the water. If all four sides had been furnished with steps, the water could not be reached vertically from the top edge of the *kuṇḍa*. It is, however, not necessary for an entire side to be vertical to allow water to be lifted up; this is as just pointed out, an unstable construction. More frequently, therefore, only a relatively small protruding vertical wall with a well-head is found inserted into one of the stepped sides of a *kuṇḍa*.

This feature seems to have derived from much simpler versions of *kuṇḍas* which were more closely related to wells. In rural areas in India, one often finds large wells, similar to small deep tanks, which have not been firmly constructed and are mainly *kaccā*. Because people have to step close to the unsecured edge of these well-tanks to lower their buckets, and because the earth easily breaks away, projecting platforms were constructed several metres down the side of such structures in a similar manner as those found in *kuṇḍas*. These simple constructions were built in stone or brick on one side (Plate 169), but at a later date the sides of such large wells might also be clad in stone. An example of a well which has been firmly constructed in stone and furnished with such a projecting platform can be seen in the fields outside Dhar (Dhār) in western Madhya Pradesh. *Kuṇḍas* having such large projecting buttresses include for example the *kuṇḍa* at Pipad (Pīpāḍ) in Rajasthan, the Bhīm Lāt Kuṇḍa in Chitorgarh, and the Ujālā Bāvlī at Mandu (Plate 166). The late tenth-century *kuṇḍa* at Vasantgarh (Vasantgaṛh) in Rajasthan, and the probably even earlier Zīl Mīl Bāvlī at Pachrahi (Pacrāī) in Madhya Pradesh (Plate 171), have an interesting variation of this feature. Both *kuṇḍas* have two projections next to each other, and between them, supported on the protruding buttresses, is a Persian wheel turned by bullocks to lift the water. In *kuṇḍas*, the feature of the projecting central wall sections was developed further and often lost its former purely practical function as a secure platform to raise water. The projecting sections of the *kuṇḍa* walls, which are of the same height as the funnel-shaped basins, and typically protrude from the centre of usually only one side, will be referred to as 'risalits' (Fig. 23);[25] this is an Italian term used mainly to describe the tendency of certain parts of the façades of European palace and church architecture to protrude forward from a vertical wall.[26]

At Bundi, there are two identical deep *kuṇḍas* on either side of the road leading to the Chogan Gate. The Nagar and the Sāgar Kuṇḍas have one shallow risalit each in the centre of their northern sides (Plate 172). These sides are almost vertical walls with only narrow terraces and no steps. The three other sides, although they have pyramidal blocks of short steps, are also extreme-

[25] P. Reclam, 1996, *Kleines Wörterbuch der Architektur*, pp. 109, 116; W. Müller & G. Vogel, 1989, *dtv-Atlas zur Baukunst*, vol. 2, pp. 427, 457, 465, 469, 485, 495.

[26] M. A. Dhaky refers to these protruding wall sections once as *pratolī* (elsewhere spelled either *pratolī* or *pratoḷī*), meaning a 'gatehouse' (Dhaky & Meister, 1991, pp. 352, 457), and on the photographs of the Vasantgaṛh Kuṇḍa in the AIIS photographic archive, Gurgaon, as *aṭṭālikā*, meaning a house of two or more storeys, or a lofty house palace (translations from: P. K. Acharya, 1993, *An Encyclopaedia of Hindu Architecture*, Manasara Series, vol. 7, p. 12).

ly steep. In both *kuṇḍas*, the façades of the protruding risalits have been decorated with string courses and small niches containing Hindu religious images. The top section of the risalit where the well-head is located consists of a kind of throne platform supported by elaborately carved corbelled brackets which protrude slightly further over the water. The backs of the platforms have been beautifully carved with lotus designs. Within the City Palace at Bundi, in the formal garden in front of the Citra Mahal, there is a small *kuṇḍa* which has four such throne platforms in the centre of each of its four sides. A particularly beautifully adorned example of a central risalit can be seen in the *kuṇḍa* in the village of Kamalapuram, south of the Royal Centre at Vijayanagara. Four brackets, two carved as leaping lions, and two as corbels with lotus pendants (*puṣpabodigai*) protrude from its front, and its sides are carved with Hindu deities, the sun and the moon, *mithuna* couples, snakes and other animals in low relief (Plate 173). The *kuṇḍa* on the main road in Eklingji has a central risalit about ten metres wide. It is decorated with string courses and has two large corbelled openings one set above the other in its centre. The niches enable one to see into the well shaft which is contained within the projecting risalit on the west side of the *kuṇḍa* (Plate 174).[27] The *kuṇḍa* at Eklingji constitutes an important mediating stage in the development of the risalits. It is related to the examples mentioned above, where the projecting risalits were solid and only decorated with shallow niches, but here the central risalit has been hollowed out and the corbelled openings open up the well shaft behind. This prefigures later developments, where the solid mass of the risalit would be carved out to contain galleried apartments and shrines (Fig. 23 b).

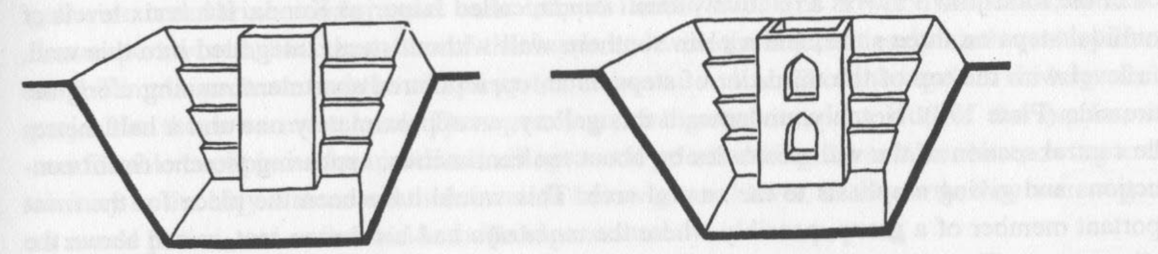

Fig. 23. *Kuṇḍas* with risalits

The simplest version of this subsequent stage in the development can be seen in the Sundhārā in the royal garden behind the palace at Patan in Nepal (Plate 175). On the north side of the *kuṇḍa* is a complex risalit. It contains a small room with a balcony in front and a platform above, which is similar to the ones in the Nagar and Sāgar Kuṇḍas; it also has beautiful carvings at the back. The sixteenth-century Pannā Miyāṃ Kuṇḍa in Amber village has a wide risalit on its south-western side.[28] This particular risalit not only juts forward into the *kuṇḍa* but extends backward to double the surface area of the platform. Above is a large square platform, surrounded by a low ornamental fence, overlooking the water basin. The upper part of the risalit is solid and has two elegant well brackets. Underneath a pronounced *chajjā*, the risalit has an apartment for the women, who

[27] The location and purpose of these well shafts within the structures of the *kuṇḍas* have been discussed earlier in this chapter.

[28] This is one of the few *kuṇḍas* which are not oriented towards the cardinal points of directions.

while the men were bathing would remain hidden by a green stone screen (*jālī*).[29] An interesting arrangement, which has two protruding risalits next to each other, is the Sabirna-dha-kā Kuṇḍa at Bundi. This design seems to have derived from examples similar to the *kuṇḍa* at Vasantgarh and the Zīl Mīl Bāvlī at Pachrahi, discussed above. In the Sabirna-dha-kā Kuṇḍa, the space between the two projecting blocks is occupied by a galleried apartment on the top level, and by pyramidal steps on the two levels below it (Plate 176). A further open apartment was set on top, towering above the upper edge of the *kuṇḍa*-basin. Although sections of the large risalit are carved out, it still retains a character of substantial solidity. In the Mandākinī Kuṇḍa at Bijolia, the entire risalit is hollow and transformed into an open balcony. A closed part remains only at the back, containing a shaded room behind the lavishly decorated open porch facing the water of the *kuṇḍa* (Plate 177). At high water level, when the porch is almost entirely submerged, its roof can still be used for drawing up water. An even more skeletal construction, though still reminiscent of earlier solid risalits, exists on the west side of a *kuṇḍa* near the so-called Underground Temple at Vijayanagara. In many of these cases, the function of the risalits has changed. Originally, they were designed as places from which water could be securely pulled up from a *kuṇḍa*, but they developed into cool retreats in the summer heat and were often used as pleasure resorts for royalty.

More closely related to *kuṇḍas* with three stepped sides and a plain vertical wall, but continuing the tendency to break into one side of a *kuṇḍa*, as described with the risalits, are instances where apartments have been constructed behind the plain wall at one side of the *kuṇḍa*. At Bundi, west of the Rānī-jī-kī Vāv[30] is a relatively small *kuṇḍa*, called Jaipuriyā Kuṇḍa. It has six levels of pyramidal steps on three sides, and a plain southern wall without steps. Integrated into this wall, on a level with the top of the third tier of steps, is an open pillared apartment running along the entire side (Plate 178). Notably, underneath this gallery, an approximately one and a half metres wide central section of the wall protrudes by about ten centimetres, appearing to echo risalit constructions and giving emphasis to the central arch. This would have been the place for the most important member of a group, possibly where the *mahārāja* had his throne seat, raised above the cooling water. The sixteenth-century *kuṇḍa* at Sabli in Gujarat, which is part of an Islamic pleasure garden complex, has apartments on two levels integrated into the west side. Similar in style to the open balcony of the Mandākinī Kuṇḍa at Bijolia, the balconies, balustrades and doors at Sabli are beautifully carved and decorated. Apartments, or at least deep niches on three levels, were constructed on the south side of the Ujālā Bāvlī at Mandu, and a fourth level was added through a small domed pavilion which reaches above the upper limit of the perimeter wall (Plate 170). The *kuṇḍa* in the complex of the Quṭb Shāhī Tombs at Golconda has an open arched gallery running almost around all four sides of the basin. A large pyramidal staircase is to be found underneath the gallery on the west. Although the central part on the north side does not protrude as the risalits discussed above, it is marked by a large pointed arch underneath the gallery, and a second apartment, three arches wide, protrudes above the upper perimeter (Plate 179). Above this

[29] Livingston, 1995. p. 13.

[30] The Rānī-jī-kī Vāv is the best known stepwell at Bundi and will be discussed in the section on stepwell architecture in Chapter Six.

highest level of the construction are two sets of well brackets with pulley mechanisms. In the monumental Potarā Kuṇḍa, next to the Kṛṣṇajanmāṣṭamī (birthplace of Kṛṣṇa) at Mathura (circa 95 metres square), apartments run almost around the entire *kuṇḍa*, interrupted by a wide ramp on the north, and broad flights of steps centrally located on the other sides. There is a large protruding risalit with a pulley mechanism on top and three levels of apartments in the south-eastern corner. The basin was furnished with steps by Mahādjī of Sindhiyā in 1782 AD, and its name derives from Kṛṣṇa's infant napkins (*potarā*) which Devakī is said to have washed in the *kuṇḍa*.[31]

The most elaborate and complicated arrangements of this kind were constructed at Osian and at Abaneri in the late eighth and the early ninth century respectively. The *kuṇḍa* at Osian has eight tiers of steep pyramidal steps. Unfortunately, the apartments which once must have covered the entire east side, and which were connected and approached by passages from behind, are quite dilapidated (Plate 180) and the top chambers have been completely destroyed. From the centre of this side still protrudes a massive risalit which also contains the well feeding the *kuṇḍa*. Because the style of the columns and carvings remaining *in situ* on all levels of the east side are contemporary with the other parts of the *kuṇḍa* and the temples around, this side seems originally to have been designed in this elaborate way. Because the Ābānerī Kuṇḍa, also called the Chānda Bāvlī, at Abaneri has Mughal Period pavilions at its upper level, and was rebuilt during the eighteenth century,[32] it conveys a better picture of how both *kuṇḍas* may once have looked (Plate 21). At Abaneri, there are about eleven levels of triangular blocks of steps, and about six levels of galleries and shrine rooms were built into its north side. The lower two levels, which are contemporary with the other sides of the *kuṇḍa* and with the adjacent early ninth-century Harṣatmātā Temple, protrude and the central part contains the well shaft of the *kuṇḍa*. Two large shrines project at the lowermost level. The apartments, integrated into one side of the *kuṇḍas*, were comfortable cool places for travellers, villagers and priests during the summer heat; owing to their vicinity to the holy waters and their location below ground level, in the earth, they were regarded as sacred places, and as thresholds to the supernatural.

Linear Kuṇḍas

A further sub-group of *kuṇḍas* is more linear in nature and layout, whilst the ones discussed before have been mainly square and concentric. Linear *kuṇḍas* are found in hard or rocky soil where their side walls can be almost vertical because the ground is stable and cannot easily slip away. Although the sides are usually still furnished with small parallel steps to secure the structure, the steps tend to be so small, and the angle of the sides so steep, that one cannot normally use them to descend to the water. Steps permitting access to the bottom of the pit are only present on one side of the *kuṇḍa*. This side with the flight of steps has to be far more shallow than the three other sides, and therefore creates a long and deep open cut in the ground leading to the *kuṇḍa*-basin. In order to buttress the steep sides of the basin, these constructions were often

[31] F. S. Growse, 1979, *Mathurā: A District Memoir*, p. 131; Entwistle, 1987, pp. 211, 319.

[32] Meister & Dhaky, 1991, p. 237.

built in a staggered way, reducing the width of the *kuṇḍa*-basin to the size of a narrow flight of steps (Fig. 24). Because of their shared linear nature, this sub-group of *kuṇḍas* is related to stepwells. The differences between a linear *kuṇḍa* and a stepwell are that wells usually go down

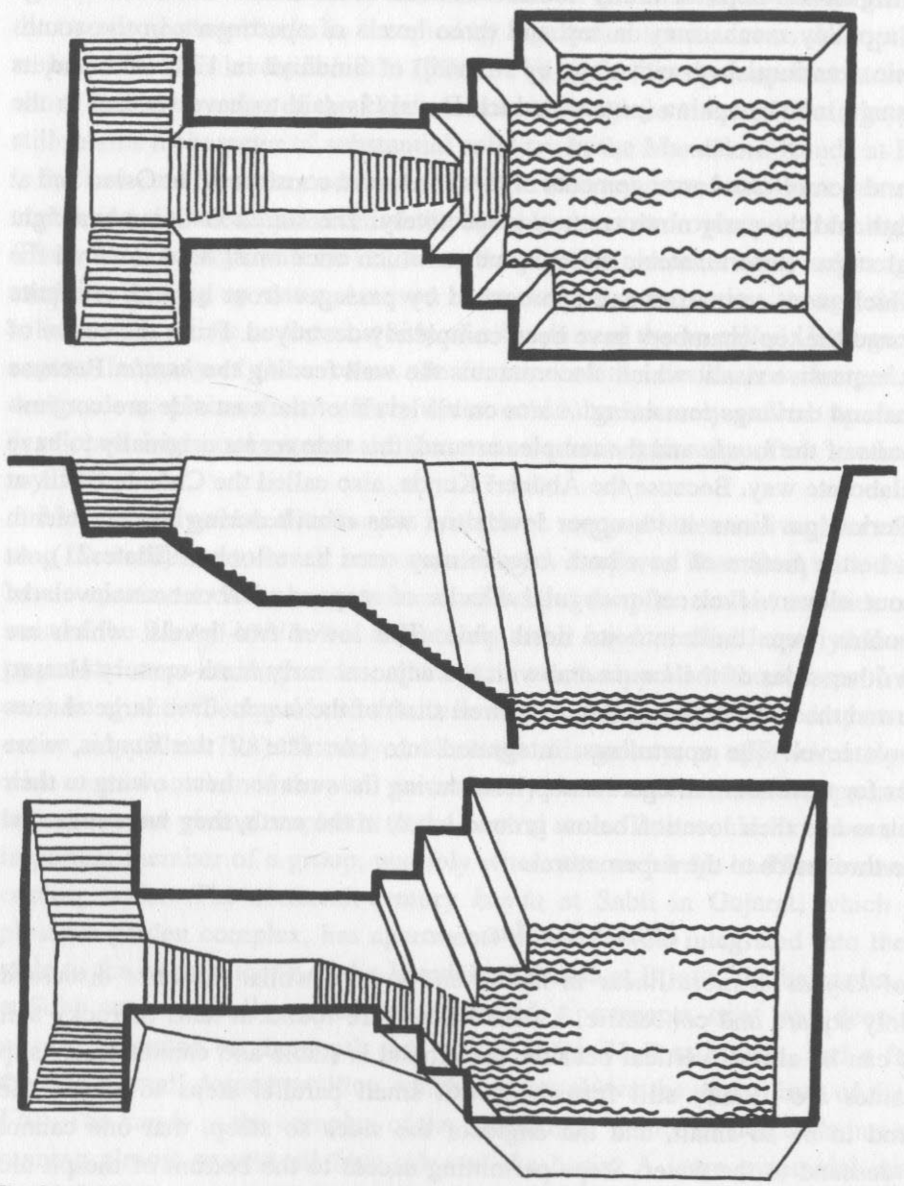

Fig. 24. Schematic drawing of a linear *kuṇḍa*

further, and therefore have longer and narrower steps; and that while in a linear *kuṇḍa*, the steps make up the fourth side of a rectangular basin with a continuous transition from the basin to the access steps, in a stepwell there is a clear distinction between the circular well shaft and the linear flight of steps (Fig. 30). Moreover, the water basins of *kuṇḍas* are larger than well shafts, and

although the sides of a *kuṇḍa*-basin are very steep, they always widen towards the top, while wells usually have vertical walls.[33] As a consequence, linear *kuṇḍas* do not need supporting constructions against the inward thrust from the side walls.

Due to the hard and rocky soil conditions in most parts of the Deccan, and particularly in the modern state of Karnataka, this region has a substantial number of clear linear *kuṇḍas*. A large and elaborate example is located behind the Mahādeva Temple at Ittagi (Plate 20). To the north, it has a circa fourteen metres square *kuṇḍa*-basin with steep sides of narrowly stepped stone. The basin tapers towards the south and goes over into a flight of steps which leads up to a terrace where the steps split into two side stairs pointing east and west, thus shortening an otherwise long flight of steps. The length of the steps is equal to the sides of the basin. A smaller but steeper example is located in the compound of the mid-twelfth-century Jaina Basti at Halebid (Haḷebīḍ) in Karnataka. Today, the upper parts of the wall of the *kuṇḍa*-basin have been rebuilt in rubble, but earlier photographs by G. Verardi still show the original walls.[34] A similarly deep *kuṇḍa* with very steep sides but no flight of steps, is located in Nandi village in Karnataka, in the compound of the Bhōgaṇandīśvara Temple. At Halebid, in order to shorten the long flight of steps, they turn from east to south after a small terrace half way up the climb. A particularly impressive and complicated example is the well-preserved eleventh- to twelfth-century linear *kuṇḍa* in front of the small Manikeśvara Temple at Lakkundi (Plate 184). The *kuṇḍa*-basin to the north measures about sixteen metres square, and the long flight of steps to the south is of equal length. In this example, there are two more flights of steps leading down in the centre of the east and west sides of the basin, but they can only be used to reach the water at high water level, when the basin is filled to almost three quarters of its height, at which point a wider terrace runs around the *kuṇḍa* (Plate 181). At this level, eight small shrines, now empty, were integrated into three sides of the basin, and additional shrines were placed at intervals down the three flights of steps. The most unusual feature of this complicated construction, however, is the wide bridge which leads over the long flight of steps in the south towards the temple in the west. The bridge is supported by a construction of pillars and beams underneath. Although this arrangement is strongly reminiscent of stepwells, where supporting constructions of posts and beams are needed to counterbalance the thrust of the vertical side walls, the walls at Lakkundi are horizontally stepped, and widen towards the top. Here, therefore, the bridging construction was not needed for strengthening purposes. It seems to have been inserted at a later stage, to be an aesthetic choice and also a ritual requirement. At Lakkundi, the water structure is many times larger and more elaborate than the small and relatively simple adjacent temple, making a strong statement about the importance of water and its religious connotations.

[33] Only the sub-group of open stepwells, discussed in the following chapter, Chapter Six on wells, widens slightly towards the top. But even these wells are clearly distinguished from linear *kuṇḍas* by having only few large recessing tiers, each an entire storey in height (while linear *kuṇḍas* widen towards the top in many small steps), and even more clearly by having a separating device between the well shaft and the stepped corridor, which is not found in straight *kuṇḍas*.

[34] G. Verardi, 1980, "La Bāolī Hoysala Di Huligere", plate VI.

There are many examples of smaller versions of this sub-group, as for instance the *kuṇḍa* in the walled temple enclosure of the Navaliṅga Temple complex at Kukknur (Kukkanūr) in Karnataka. The structure dates from the late ninth or early tenth century AD and tapers to the west, where a flight of steps leads up to a platform and then turns south. Although this *kuṇḍa* is relatively small (about 9 by 10.5 metres), it is not very deep, and could therefore have easily been furnished with proper steps on all sides; but the same aesthetic of a linear approach was chosen which required slightly less building material for the walls. In the *kuṇḍa* of the twelfth-century Cenna Keśava Temple at Belur (Plate 24), the difference between the stepped side walls and the proper flight of steps in the west is visually less pronounced. However, the "treads" of the walls offer so little foothold that in practice they would not be confused with the steps proper.[35] A further example is the fifteenth- to sixteenth-century Joḍ Bāvlī, also called Joḍ Gumbaz-kī Bāvlī because it is situated close to the tomb of Khān Muḥammad, at Bijapur. The *kuṇḍa* in the village of Kamalapuram near Vijayanagara, already mentioned under the former sub-group for its beautifully decorated central risalit, also follows the typical plan of the linear *kuṇḍas* by tapering towards the east. It is quite shallow and its steps can be descended on all four sides.

Not all *kuṇḍas* in the Deccan, however, fall into this sub-group, and other examples such as the large circa seventeenth-century *kuṇḍa* at Mudabidri, or the small example in the Ibrahim Rauzā at Bijapur, follow the most simple version of *kuṇḍas* with steps or large tiers on four sides. Examples of linear *kuṇḍas* can be found outside the Deccan in other parts of India, where they usually follow a layout similar to those at Kukknur, Belur or Kamalapuram, by not being so deep but having a pronounced tapering end. Good examples include Lakṣmī Kuṇḍa at Benares whose shape tapers towards the east and leads up to a flight of steps, and the *kuṇḍa* in front of Daryā Khān's Tomb at Mandu, which tapers to the south. The Kuṭṭam Pokuna (Twin Pools) at Anuradhapura are rectangular without a tapered end, but they are very deep and have the typical steep walls which are stepped to secure the ground but are impossible to descend (Plate 196). Proper stairs with wider steps are provided at both ends of the *kuṇḍas* and on the western side of the southern *kuṇḍa*. A similar arrangement is found in the *kuṇḍa* behind the Pāpanāśi Temple at Bhubaneshwar. A particularly deep and impressive *kuṇḍa* of this sub-group is the Jhālrā Kuṇḍa in the Dargāh at Ajmer (Plate 182). It is comparable in depth and steepness to the *kuṇḍas* at Ittagi and Lakkundi and its sides are made up of narrow stone steps.

Conduit-Kuṇḍas

A third sub-group of *kuṇḍas* has large tiers on all four sides but proper steps, usually positioned centrally only on one side. These *kuṇḍas* are multi-stepped and often very deep. The most important characteristic of this group of *kuṇḍa* is that the water issues out of water spouts in the lowermost retaining wall of the structures, and that the overflow is channelled out of those basins which do not usually contain or store water.

[35] The steps in the west measure 37cm (tread) and 27cm (riser); the stepped sides measure respectively 16cm and 30cm.

The conduit sub-group of *kuṇḍa* is the most common water structure of Nepal, locally called *ḍuṅge-dhārā*, *gaihrī-dhārā* (Nepali), *gā-hiṭī* or *lavahaṃ-hiṭī* (Newari) (Plate 185).[36] The *dhārās* tap shallow ground water aquifers directly and distribute the water through spouts. As the ground water table does not vary much between the seasons and on the slopes of the Himalayas water is available all year round, these *kuṇḍas* do not have to store water. The *dhārās* provide continuous fresh running subsurface water which is clean and more desirable than stagnant water.[37] The overflow of some *dhārās* is channelled into storage basins (*pokharīs*) while others distribute their water into the rivers. The oldest *dhārās* surviving in Nepal date from the sixth century AD, but most of them have been extensively rebuilt. Usually only the spouts, which are normally carved or cast in the forms of mythical animals or other sculptural elements, and which bear the dated inscriptions, still date from this early period. Although most *dhārās* are public water structures found in the squares and streets of the towns, conduit-*kuṇḍas* were also constructed inside the palaces of the Malla kings (ruled 1200-1769 AD) (Plate 264).[38] An unusual example of this sub-group from Karnataka is the circa eleventh-century *kuṇḍa* near the Sūḷe Basti at Humcha (Humca) (Plate 183). It is a relatively shallow structure with a stone water conduit in the centre of its western side. Central flights of steps lead down into the *kuṇḍa* on the other sides, and the walls are decorated with ornamental flowers and small elephant sculptures adorn the steps. There are also several examples of *kuṇḍas* in Sri Lanka which are fed by decorated water spouts. The difference is that these structures usually store water. The northern basin of the eighth- to ninth-century Kuṭṭam Pokuna at Anuradhapura, is fed by a stone water conduit which, though severely damaged, still resembles the shape of a *makara*. The twin *kuṇḍas* were used as ritual baths by the monks. The Kumāra Pokuna at Polonnaruva has two stone water spouts flanking the stairs on the western side, carved in the shape of crocodiles or mythical *makaras* (Plate 187). These are further examples of the remarkable connection between the water architecture of Nepal and Sri Lanka, mentioned earlier.

Examples Relating to other Types

Having both a tank in which water is stored, and in most cases also a well, *kuṇḍas* lend themselves particularly well to creating hybrid forms where the inherent aspects of tanks or wells may be pronounced more strongly. An example where the well element of a *kuṇḍa* has been thus emphasised is the small *kuṇḍa* north-east of the Kuśavart Tīrtha at Trimbak. The *kuṇḍa* measures about nine metres square and has three parallel steps on all four sides, and a fourth step on the top of the basin, in the centre of each side. The square basin contains a central octagonal well shaft which becomes circular lower down the shaft. The sequence from a square over an octagon to a cylindrical shape creates an inverted *liṅga*. The fact that there is no perimeter wall surrounding

[36] Whilst *ḍuṅge-dhārā* and *lavahaṃ-hiṭī* both mean stone water conduit, *gaihrī-dhārā* and *gā-hiṭī* both mean deep water conduit.

[37] A small number of *dhārās* in the Kathmandu Valley have a small additional circular well shaft in one of their corners, as for example the Kva Hiṭī at Kathmandu.

[38] For more information on the water architecture of the Kathmandu Valley see J. A. B. Hegewald, "Water Architecture of the Kathmandu Valley: Function and Faith", 1997(a).

the structure and that there is no pulley mechanism to draw up water, both typical features of wells, makes the structure still more clearly a *kuṇḍa,* although the funnel-shaped basin is small in contrast to the well shaft. A further *kuṇḍa*-well hybrid is the Gaṅguā Kuṇḍa, also called the Gaṅguo Tank, at Loteshvar (Loṭeśvara) in Gujarat. It has a central circular well shaft which reaches up to the level of the surrounding ground and is surrounded by four *kuṇḍa*-basins. The basins are connected to the central well, and are fed by it through openings in its side wall (Fig. 25). An example of a *kuṇḍa* relating to stepwell constructions is the Ujālā Bāvlī at Mandu. This example has pyramidal steps on its south and west sides, but because the steps are very steep and narrow, a longer and shallower flight of steps has been constructed leading down to the water in the western corner of the northern side (Plate 166). This wedge-shaped flight of stairs is strongly reminiscent of stepwells.

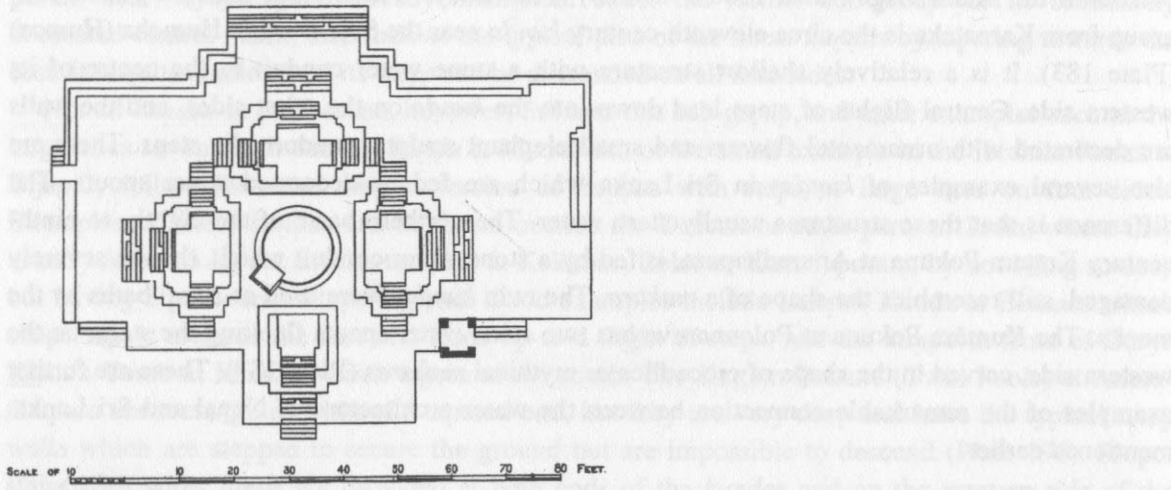

Fig. 25. Loteshvar: Gaṅguā Kuṇḍa (Burgess & Cousens, 1903)

The Lolārka Kuṇḍa at Benares has three long flights of steps leading down to a square *kuṇḍa*-basin at the bottom (Fig. 26). A narrow flight of steep steps leads to a central deep tank with vertical sides which enabled the builders to descend quickly to a considerable depth within a smaller area and therefore more economically than if the whole basin had been furnished with long tiers of steps. At the bottom of the narrow stairs is a typical funnel-shaped *kuṇḍa*-basin and the well feeding the basin is found next to it, on the east. In this *kuṇḍa,* the interconnection between the basin and the well shaft in the shape of a tall open pointed arch is clearly visible (Plate 186). The Lolārka Kuṇḍa is believed to be one of the oldest sacred sites of Benares and is dedicated to the sun god Sūrya. Annually, in the month of *Bhādrapada* (August–September), a large religious fair is held at the *kuṇḍa,* and married couples come to bathe in the water to gain male offspring.[39]

[39] D. Eck, 1993, *Banaras: City of Light*, pp. 177–178; N. Gutschow & A. Michaels, 1993, *Benares: Tempel und religiöses Leben in der heiligen Stadt der Hindus*, p. 77.

The *kuṇḍas* at Bhinmal (circa eighth century AD and later) and at Umta (circa late tenth century AD) in Gujarat both have funnel-shaped *kuṇḍa*-basins with pyramidal step formations, and do not have long and narrow flights of parallel steps; nonetheless they resemble stepwell constructions because their fourth walls connecting the *kuṇḍa*-basins with the well shafts behind have been opened up, and in shape are similar to the supporting constructions commonly used in stepwells to withstand the substantial side thrust evident at such depth (Plate 188).[40] Many *kuṇḍas* have plain berm walls straight below ground level forming an upper perimeter wall around the deep stepped basins. This feature is typical of Rajasthan, and may be seen in the *kuṇḍa* at Ghanerao, the Khatan Bāvlī in Chitorgarh, and the so-called 'Later Kuṇḍa' at Ahar (Plates 152, 192). Examples from other regions are the *kuṇḍas* at Mukhed in Maharashtra, and in the complex of the Suttur Maṭh outside Mysore (Maisūr) in Karnataka (Plate 158).[41] As a result such basins can easily be mistaken for tanks with straight walls at high water level when the funnel-shaped main part of the *kuṇḍa* is entirely covered by water. The Sabirna-dha-kā Kuṇḍa at Bundi even has three tiers of berm walls set on top of one another, before the usual pyramidal steps start their descent on all four sides.

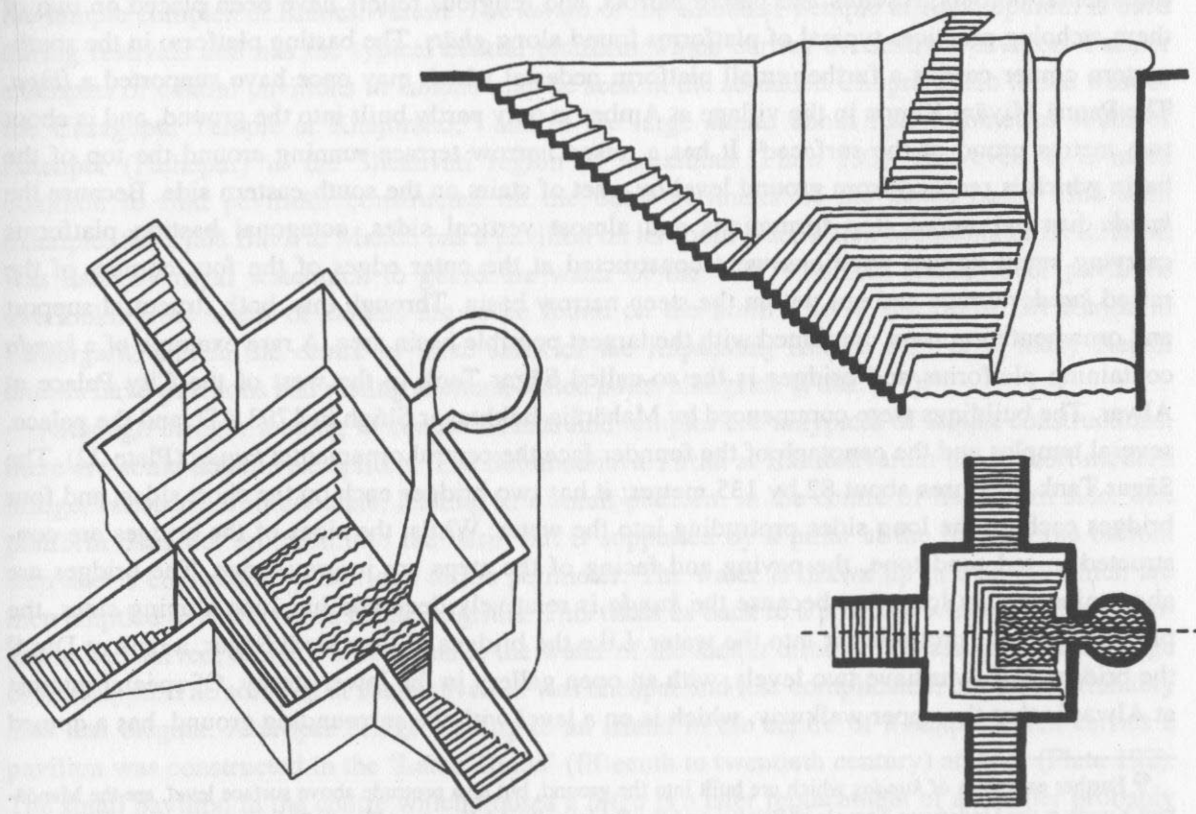

Fig. 26. Schematised drawing of the Lolārka Kuṇḍa, Benares

[40] J. Jain-Neubauer remarks on hybrids between *kuṇḍas* and stepwells in Rajasthan. She calls them either *kuṇḍa-vāpī* or *kuṇḍa-vāv* which she translates as 'stepwell pond' (*The Stepwells of Gujarat in Art-Historical Perspective*, p. xiii).

[41] The Suttur Maṭh is a Liṅgāyat institution.

3. PLATFORMS, TEMPLES AND IMAGES IN KUṆḌAS

Bastion platforms, which have been noticed in relation to *ghāṭs* and tanks, are also associated with *kuṇḍas*. Because *kuṇḍas* often have very steep sides and a relatively small basin at the bottom of the steps, the builders had to be inventive to integrate these additional structures into the narrow funnel-shaped basins. The Viśvanātha Kuṇḍa at Ellora, which is also called Brahmā Kuṇḍa or Aṣṭha Tīrtha, because eight heavenly rivers are believed to flow into the basin, has four platforms in its corners. The *kuṇḍa* has a high berm wall with empty niches, and is penetrated by steps in the centre of all four sides. Below the berm wall, the parallel steps are arranged in four large tiers. Because eight small shrines were constructed around the water basin at the bottom of the funnel, the bastion platforms were integrated into the corners of the uppermost tier of steps, furthest away from the water (Plate 189). The purpose of bastion platforms in their early development along river banks was to protect the edges of *ghāṭs* against the force of running water. However, the platforms in the Viśvanātha Kuṇḍa, though still massive octagonal structures, have been used as decorative features, located away from the water. Their sides are beautifully carved with ornamental borders, rosettes and paired parrots, and religious reliefs have been placed on two of them, echoing practices typical of platforms found along *ghāṭs*. The bastion platform in the south-eastern corner carries a further small platform pedestal which may once have supported a *liṅga*. The Pannā Miyāṃ Kuṇḍa in the village at Amber is only partly built into the ground, and is about two metres proud of the surface.[42] It has a raised narrow terrace running around the top of the basin which is reached from ground level by a set of stairs on the south-eastern side. Because the *kuṇḍa* has pyramidal step formations and almost vertical sides, octagonal bastion platforms carrying small domed pavilions were constructed at the outer edges of the four corners of the raised *kuṇḍa*-terrace and not within the steep narrow basin. Through this, both structural support and ornamentation were combined with the largest possible basin area. A rare example of a *kuṇḍa* containing platforms and bridges is the so-called Sāgar Tank to the west of the City Palace at Alwar. The buildings were commenced by Mahārāja Bakhtavar Siṅgh in 1793 AD, and the palace, several temples and the cenotaph of the founder face the central ornamental *kuṇḍa* (Plate 22). The Sāgar Tank measures about 82 by 135 metres; it has two bridges each on the short sides, and four bridges each on the long sides protruding into the water. Whilst the sides of the bridges are constructed in red sandstone, the paving and facing of the steps are in grey stone. The bridges are about nine metres long, but because the *kuṇḍa* is relatively deep and has long sloping sides, the platforms do not protrude far into the water. Like the bridges in the Gopāl Sāgar, a tank at Dig,[43] the bridges at Alwar have two levels, with an open gallery in the lower storey. Of special interest at Alwar is that the upper walkway, which is on a level with the surrounding ground, has a domed pavilion built onto the octagonal platform demonstrating the dual purpose design of such double bridges even more emphatically than the example in the Gopāl Sāgar. Moreover, as can be seen from the step formations at Alwar, the platforms were planned to extend higher above the water, whilst the aim of the design at Dig was to bring the construction as close to the water surface as possible. Here, because the bastion platform underneath the gallery and the upper pavilion are comparatively high, the impression of a three-storeyed construction is conveyed. At Alwar, these elaborate bridge-constructions could only be accommodated in the *kuṇḍa*, because the basin is comparatively large and shallow, and is as such also related to tank architecture.

[42] Further examples of *kuṇḍas* which are built into the ground, but also protrude above surface level, are the Mandākinī Kuṇḍa at the Undeśvara Temple at Bijolia, and the Sabirna-dha-kā Kuṇḍa at Bundi. The small *kuṇḍa* in the Ibrahim Rauzā at Bijapur is very shallow and does not seem to go deeper than the height of the raised platform which also carries the tomb and the mosque of the funerary complex. This feature is not unique to *kuṇḍas* and can also be encountered in tank architecture, as for example in the small tank at Ramkund (Rāmkuṇḍa) near Lodruva (Ludravā) in the desert Thar in Rajasthan.

[43] The Gopāl Sāgar at Dig was discussed in the last chapter, on tanks.

Platforms may also be found in the centre of *kuṇḍas* with solid floors, where there is no well shaft or where it is located on one side. The Kumāra Pokuna at Polonnaruva has a circular platform carved as a lotus in its centre on which the king allegedly used to sit surrounded by water (Plate 190). The Sundhārā in the garden of the Royal Palace at Patan in Nepal, has a central platform carrying a lotus bud pillar (Plate 175). Such pillars are typical of Islamic architecture, and the *kuṇḍa* of the Ibrahim Rauzā at Bijapur also has an almost identical platform with a lotus pillar in its centre. A plain lamp pillar is to be found in the centre of the Setumadhava Tīrtha in the temple complex at Rameshvaram. The *kuṇḍa* of the Kāmākṣī Temple at Kanchipuram is used during festivals and has the typical central platform which carries the festival pavilion. Further examples of central pavilions in *kuṇḍas* may be seen in the so-called Choprā Tālāb to the west of the Citragupta Temple at Khajuraho,[44] and in the large *kuṇḍa* about four kilometres south of Fatehpur (Faṭhepur) in the Shekavati region of Rajasthan (Plate 193). However, it is more common to find pavilions constructed on the outer perimeter of the *kuṇḍa*-basin. One such example, the Ujālā Bāvlī at Mandu has a pavilion on its south side, which according to D. R. Patil, was used by royal watchmen to guard the water of the well.[45] Further examples of pavilions overlooking the water of *kuṇḍas* are to be found on the north side of the Bhīm Lāt Kuṇḍa in Chitorgarh, and in the centre of three sides of the Kāpaḍvañj Kuṇḍa. Similarly, many Nepali *dhārās* have pavilions and resting shelters, called *pāṭis*, alongside (Plate 191).[46]

Although bridges leading to central islands and temples are untypical of *kuṇḍa* constructions, there are some notable exceptions. The Setumadhava Tīrtha at Rameshvaram has a short modern bridge, constructed in concrete, leading to a small platform in the centre of its eastern side. The platform does not protrude into the basin but is supported by a pillar at the level of the bottom step and is connected by a bridge to the perimeter. The water is drawn up in buckets which are then emptied over the heads of the pilgrims. This takes us back to a problem which the risalit subgroup had solved, namely how to make the water of the *kuṇḍa* directly accessible from the edge of the *kuṇḍa*. The solution at Rameshvaram was cheaper and less complicated, but unquestionably also less elegant. A proper bridge leading to an island in the centre of a *kuṇḍa* which carries a pavilion was constructed in the 'Later Kuṇḍa' (fifteenth to twentieth century) at Ahar (Plate 192). The small pavilion in the centre which houses a *liṅga* is a later replacement of an earlier probably

[44] E. Zannas, 1960, *Khajuraho*, pp. 108, 168, plate 36.

[45] Patil, 1992, p. 29.

[46] There is an unusual example of a *pāṭi* containing a water conduit, near the Caṇḍeśvarī Temple in Bhaktapur.

larger structure.[47] The *kuṇḍa* is surrounded by a typical berm wall containing recesses for religious images and pyramidal steps below. The *kuṇḍa* must be comparatively shallow for its island to be positioned in the centre of the basin. Another notable example is the bridge found at the southern end of the *kuṇḍa* at Lakkundi (Plate 184). Because this large and linear *kuṇḍa* is located immediately in front of the Manikeśvara Temple, one would otherwise have to walk around it to reach the temple. But the bridge leading over the narrow steps provides direct access to the porch of the temple. Pilgrims thereby reach the temple by crossing water, or at least by crossing the steps of a water structure, and this arrangement gives tangible shape to the idea of the *tīrtha*, and as such can be related to bridges leading to central islands with temples, or other religious buildings in tank architecture.

Small temples were also constructed inside *kuṇḍa*-basins. In the corners of the eighth-century *kuṇḍa* at Roda, at the level of the fourth landing of pyramidal steps, are four shrines. They were built into the stepped sides of the basin, face in different directions, and their deities guard the sacred water site.[48] The shrines are found at the bottom of the deep stepped basin, which is now dry yet must originally have been under water for considerable periods of the year. At Abaneri, two shrines were built against the bottom part of the central risalit which itself contains galleries and shaded apartments at a higher level (Plate 21). The shrines were constructed so close to the water surface that it appears to be intentional that they should be partly submerged for certain periods of the year. Because water is regarded as sacred, and religious images are regularly bathed and immersed in water, it is particularly auspicious for shrines and their images to be submerged and so become imbued with the sacred power of the water. In the dry season, these temples reappear from the water and the images are worshipped by the people. The *kuṇḍa* at Dedadara, whose layout is closely related to the Roḍā Kuṇḍa, has similar shrines in the corners but outside the basin, on the upper edge of the *kuṇḍa* (Plate 165). The *kuṇḍa* in the compound of the Cenna Keśava Temple at Belur has two small temples placed on the upper edge of its basin flanking the stairs on the west side. Nearby, in the village of Huligere, is a *kuṇḍa* (circa 1142–1173 AD)[49] which is surrounded by a series of miniature temples integrated into the steep parallel steps, two-thirds of the way down to the water. There are two shrines on each side of the central stairs, and five more on each of the other sides. A similar arrangement of eight small temples surrounding the water-filled basin above the lowermost tier of steps is found in the Viśvanātha Kuṇḍa at Ellora (Plate 189); the temples contain śiva-*liṅgas*, images of Śiva, Gaṇeśa and Durgā. At Śivarātri, a major bathing festival is held in the *kuṇḍa*, and pilgrims worship at the shrines in the water basin. Although unusual for narrow *kuṇḍa*-basins, a complete *śikhara* temple with porch was built into the 'Early Kuṇḍa' at Ahar, dating from circa 1000 AD (Plate 194). Because the *kuṇḍa* has steep sides and a small basin, the temple fills it almost entirely and little remains of the water basin. The *prāsāda* of the temple rests on the edge of the *kuṇḍa* and a long open columned porch stretches

[47] An earlier photograph from the AIIS at Gurgaon (Photo No. 28.80) shows the same site without the replacement pavilion, with only the central island carrying the *liṅga*.

[48] The Gaṇeśa shrine faces north-east, the Sūrya shrine faces south-east, the shrine of Viṣṇu faces south-west and that of Pārvatī north-west.

[49] Verardi, 1980, p. 108.

out over the basin itself. Underneath, it is supported by further columns which rest on a lower floor set into the water of the basin. By being double-storeyed, the temple can adapt to changes in water level, the effect of which shows more clearly in a small basin than in a tank with a large surface area.

Although examples which illustrate distinct design themes, such as bridges, islands and temples in water, have previously been discussed in connection with *ghāṭs* and particularly with tanks, and have been shown to feature in *kuṇḍa* constructions, the latter must be considered as exceptions. Tanks and *kuṇḍas* both contain water and are related to each other, but they are distinct types of water basins. While tanks are generally shallow horizontal structures providing ample space and ground support for platforms, bridges, islands and temples, *kuṇḍas* usually have greater depth, are more compact constructions, and vertical in layout. Space for experimentation with additional building elements in *kuṇḍas* was gained in the deep and extended stepped sides. As a consequence, structures such as temples and pavilions were generally positioned in the vertical rather than the horizontal plane. They were integrated either into one side (Plate 23) or set on the edge of *kuṇḍas*, but not usually into the water of the small basins. The incorporation of apartments and shrines in one side of a *kuṇḍa* has already been mentioned as typical of the risalit sub-group of *kuṇḍas*, where a vertical wall or protruding risalit was transformed into open, often multi-storeyed apartments.

The positioning of images within *kuṇḍas* illustrates two distinct religious aspects of this type of sacred water structure. On the one hand, the *kuṇḍa* is a point of entry for sacred water and spiritual elements from the subterranean world which should be "greeted" and honoured. This is frequently done by placing sculptures and figural friezes close to the water's surface. As such decorations are aimed at pleasing the "gods" in the water, and not at the people using the structures, these images are often hardly visible from above. On the other hand, the presence of spiritual elements in the water is given tangible form in religious sculptures adorning the sides of *kuṇḍas*. These artistic representations are aimed at the people using the water structures, and are intended for worship and veneration. The appearance and disappearance of such images as the water level varies serve to heighten the sense of the images being of "another world". Generally, only reliefs or small shrine niches containing images were integrated into the side walls of *kuṇḍa*-basins. It is especially typical of *kuṇḍas* of the conduit sub-group to have small shrines with religious images from a *śaivite*, *viṣṇavite* or one of the Buddhists religious sects above their spouts set into the lowermost retaining wall.[50] These images are actively worshipped by the people using the *kuṇḍa*. Connected with other sub-groups, however, the images are frequently either so close to the surface of the water that they remain submerged for long parts of the year, or they are positioned in inaccessible positions on the straight walls of *kuṇḍas*, so that they can hardly be seen by those coming to the water place. For example in the *kuṇḍa* above the Kukadeśvara Kuṇḍa in the settlement in the fort at Chittor, an elaborate figural frieze has been integrated into the vertical wall on the west side. Because the east side is a rough rock face which is not

[50] Whilst some *kuṇḍas* are reserved for the followers of one religious group, many *dhārās* in the towns of the Kathmandu Valley contain *śaiva*, *vaiṣṇava* and Buddhist images and religious symbols, such as *caityas* and *triśūlas*, within one water structure. See for example the Cākbā Lhoṃ Hiṭī in Tangal Tol in Patan.

accessible to people and there is no terrace below from which to look up at the carvings, it is impossible to see the images clearly. The conclusion must be that it was never the intention that visitors or pilgrims should be able to read these carvings as a sacred narrative. The friezes are only accessible to the sacred waters; they are intended to please the gods who are thought to dwell within the basin, as well as to fend off evil forces from the religious site, including evil elements liable to emerge from the depths of the waters leading to the underworld.

4. POSITIONING OF KUṆḌAS AND ACCESS TO THEM

The essential role of *kuṇḍas* as a means of access to fresh ground water combined with all the powerful religious and social connotations that surround water structures has encouraged their construction in spite of the associated difficulty and expense. Consequently, unlike the ubiquitous tank, there are few multiple examples on adjacent sites. However, there are identical twin *kuṇḍas* on either side of the road leading to the Chogan Gate at Bundi, called the Nagar and the Sāgar Kuṇḍas. The basins go very deep and both seem to tap the ground water. Further examples of twin-*kuṇḍas* are the deep and steep-sided Kuṭṭam Pokuna, Twin Pools, at Anuradhapura (Plate 196). The northern basin is fed by a freshwater spring and the southern *kuṇḍa* receives the overflow. Before the spring water enters the northern *kuṇḍa*, it passes through a small settling pit to clean the water of surface dirt. In the towns of the Kathmandu Valley, one often finds two *kuṇḍas* close to each other on city squares. In common with multiple tanks, the incentive for the construction of several related *kuṇḍas* was often hierarchical or communal religious notions of purity and pollution. At Konti in Patan, there are two relatively shallow *kuṇḍas* next to the precinct of the Kumbheśvara Pagoda. The smaller Misā Hiṭī next to the compound wall of the temple has a Buddhist *caitya* in its centre and appears to be used principally by the Buddhist Newar communities, while the larger Konti Hiṭī to the west of it is Hindu, and contains a large central Śiva-*liṅga*.

Chyāsal Hiṭī, in the north-east of Patan is not open to lower castes, while everybody has access to the water of Nāya Hiṭī located south of it on the same square. The latter has a large seated Buddha image set in a casket on a pillar in the centre, which was installed in 1980 as a sign against caste restrictions.[51] In the same way, Tāpā Hiṭī in the north of Patan has two adjoining *kuṇḍa*-basins, whilst in Karnataka, the large *kuṇḍa* in front of the Āthār Mahal at Bijapur, is flanked by two smaller basins in the north and the south which, however, are shallow tanks with plain walls. Both tanks are fed by beautifully decorated water spouts at their shorter western sides. Examples of *kuṇḍas* next to and within larger water bodies have already been discussed in connection with *ghāṭs*. At Benares, the small Gaurī and Maṇikarṇikā Kuṇḍas have been built into platforms in the *ghāṭs* adjacent to the sacred Ganges, and there are three small *kuṇḍas* constructed in the water of Puṣkar Lake (Plates 85, 157, 88).[52] One example of a small *kuṇḍa*-basin on an

[51] R. O. A. Becker-Ritterspach, 1995, pp. 10–13.
[52] For a detailed discussion see Chapter Three, '*Ghats*'.

island is to be found in the large oval-shaped tank of the Tilasvaṃ Mahādeva Temple at Taleshvar.

A feature of water structures discussed in earlier chapters, repeated as an important design feature in the construction of *kuṇḍas* and subject to a distinct process of development, is the gateway. Examples of circa eleventh- and early twelfth-century ornamental gateways lead to the deep and elaborate *kuṇḍas* at Modhera and Kapadvanj (Plates 167, 195) respectively. At Modhera, the gateway forms part of a long and clear east–west axis, established by the Sūrya Temple in the west, a detached open *maṇḍapa* in front of the temple, followed by the *toraṇa*, and the large *kuṇḍa* further to the east. Likewise at Kapadvanj, a Sūrya Temple once stood behind the gateway to the east. There appears to be a particular association between temples of the sun and *kuṇḍas*, a further example of which is found in the south of Delhi at the Sūraj Kuṇḍa. These have been interpreted as a representation of the fire–water dualism which is related to the birth of the sun out of the cosmic ocean. Whilst the temples have been seen as architectural shapes of the fire altar, the *kuṇḍas* represent water altars.[53] A. W. Entwistle, in his study of the sacred area of Braj, observed that Sūrya Kuṇḍas, *kuṇḍas* dedicated to the sun, are a recurrent and specific type of that area. According to his observations, they are usually the oldest water structures in the vicinity and may be relics of the ancient Sūrya cult that flourished particularly during the first millennium AD.[54] More prominent double-storeyed gateways are to be found in the centre of each of the four sides of the *kuṇḍa* to the south of Fatehpur and of the *kuṇḍa* in the town itself.

An example of a late tenth-century AD gateway-pavilion is that between the small temple and the deep *kuṇḍa* at Vasantgarh (Plate 197). It is located in the centre of the west side and consists of four columns which start square on the bottom and develop via an octagonal section into round pillars at the top. The roof of the gateway-*maṇḍapa* has pronounced *chajjās* and the structure is integrated into a low wall surrounding the basin. The *kuṇḍa* at Vasantgarh shows particularly clearly the connection between religious and mundane uses of water structures. Whilst the east side has two narrow protruding risalits and a large wheel to pull up water for irrigating the fields and for watering animals, the west side is dominated by religious aspects, having a prominent gateway and a small temple. In this way, the water of the *kuṇḍa* serves for both ablutions and temple rites as well as for domestic and agricultural needs.

The feature of open pillared gateway-pavilions, found also along rivers and tanks, is seen to have developed further in *kuṇḍa* constructions. There are several examples where the sides of gateway pavilions have been filled in and turned into closed *maṇḍapas* or small gate-houses with two doors on opposite sides. An example of a gate-house is on the western side of the *kuṇḍa* in the compound of the Cenna Keśava Temple at Belur (Plate 24). The outer doorway resembles many shrine entrances in that it is flanked by elaborately decorated beams with images of *dvārapālas* on the bottom, and a large lotus relief carved into the floor in front of the doorway. This elaboration shows the religious importance of the *kuṇḍa*, and that the water basin was conceived as a sacred precinct which was entered through a gate-house similar to the porch of a

[53] Haus der Kulturen der Welt, 1991, *Vistāra: Die Architektur Indiens*, p. 46.
[54] Entwistle, 1987, p. 293.

temple.[55] In the same way, the large *kuṇḍa* at Abaneri is entered through a prominent gate-house, which, however, does not seem to be contemporary with the early parts of the structure: it was probably added or replaced in the eighteenth century. Both *kuṇḍas* have enclosing walls, the one at Belur measuring about 1.2 metres in height, while the wall surrounding the Ābānerī Kuṇḍa is higher, and supports an arcaded gallery which runs along the inside. A similar arcade encloses the Brahmā Kuṇḍa at Shihor (Plate 19). Today, the Ābānerī Kuṇḍa is the headquarters of the local police. Being a strong, compact building, easy to defend, it provides a modern military dimension to this ancient concept of a water structure. Because *kuṇḍas* usually have steep sides, and are much deeper than tanks, it is even more typical of *kuṇḍas* than tanks to be surrounded by walls.

Gate-houses carry their own religious significance and there are two particularly interesting examples leading down to the water of separate *kuṇḍas*, called Siṃha Tīrtha (Lion Tīrtha) Nos. 1 and 2, on the *pradakṣiṇā-patha* around the sacred mountain at Tiruvannamalai. The gateways are in the shape of large lion sculptures with doorways between the two front legs underneath the heads of the lions (Plates 198,199). A similar lion gate, leading down to a well at Gangaikonda-cholapuram was dedicated by the Zamindar of Udaiyarpalaiyam,[56] and it is probable that the gate-ways at Tiruvannamalai were also donated by him. Lion sculptures are frequently found flanking the entrances to temples, or at the base of columns, but in these *kuṇḍas*, one large lion sculpture becomes the gateway itself, and people seem to have to pass through the lions' bodies to reach the water. This is a forceful image giving visual shape to the idea that to bathe and totally immerse oneself in water is to represent one's death and rebirth. By being swallowed by a lion, the pilgrim can be reborn in the waters of the *kuṇḍas*.

It is typical of *kuṇḍas* at temples to have steep and prominent flights of steps, distinct from the surrounding stepped sides; the steps lead to the temples above. Probably the best known example of this is the Sūrya Kuṇḍa in front of the Temple of the Sun at Modhera. In the centre of the long western side, steep parallel steps in the shape of an inverted triangle lead up from the deep *kuṇḍa* to the temple (Plate 167), and visualise the spiritual ascent of the soul which may be attained through a visit to the temple. Although the temple on the east side of the Khatan Bāvlī in Chitorgarh is small, the steps leading up to it in the middle of one side are clearly defined, because the rest of the side is made up of plain vertical walls, whilst the other three sides are stepped. The monumental pyramid of steps leading up to a temple of the sun, now destroyed, on the west side of the Sūraj Kuṇḍa south of Delhi is particularly impressive. The large oval *kuṇḍa* is believed to have been constructed by the Tomar king Sūrajpāl in the tenth century AD, and its steps were repaired by Fīrūz-Shāh Tughluq in the fourteenth century.

To summarise, *kuṇḍas* are deep stepped basins which are distinguished from tanks by their steep and often pyramidal step formation, and their funnel-shaped sides. *Kuṇḍas* vary in size and elaboration and, as discussed, can be divided into three prominent sub-groups. They are found in

[55] The temple-derived religious imagery associated with the gate-house of the *kuṇḍa* at Belur reinforces the religious importance of water structures and their close connection with temples.

[56] P. Pichard (ed.), 1994, *Vingt ans après Tanjavur, Gangaikondacholapuram*, p. 149.

religious, palace and civic contexts. Although until now it has been commonly assumed that *kuṇḍas* predominantly exist in north-western India, in the regions today known as Rajasthan and Gujarat, no study of *kuṇḍas* outside this area has until now been undertaken. The conclusion of this study is that structures of this type, whilst showing slight variations within the limitations of the category, actually exist all over India, Nepal and Sri Lanka. Their emphasis of construction in the vertical plane, allied to comparably smaller basins, determined that there was less design flexibility than in tanks or along *ghāṭs* for the development of additional structures within the water. Consequently, such development was mainly restricted to the long sides of basins, with functional structures being built into the vertical side walls, and more ornamental features around the upper edges of the basins.

Because *kuṇḍas* penetrate deep into the ground, they are considered particularly sacred. Therefore, temples, shrines and religious images are commonly associated with them. With the expansion of British colonial influence in the nineteenth century it became less common to build *kuṇḍas*, partly because in the later period Western engineering techniques introduced water provision through pipes, faucets and pumps, and partly because traditional sources of financial patronage and labour were no longer available as local rulers and elites declined in influence. Whilst the building of *ghāṭs*, storage tanks, and the digging of plain draw wells continued, the construction of aesthetically pleasing, complicated, expensive and religiously significant *kuṇḍas* did not continue on any large scale. There are, however, several examples of modern architecture which make a strong reference to *kuṇḍa* constructions. A notable example is the square courtyard of the Jawahar Kala Kendra at Jaipur, designed by Charles Correa in 1986–88, which is surrounded by three tiers of pyramidal steps, typical of *kuṇḍa* constructions (Plate 287). In the centre of the open courtyard at Jaipur, where in a *kuṇḍa* a well might have been located, a large circular red sandstone disc clearly marks the place of an imaginary well shaft (Plate 288). A further modern example which makes a clear reference to *kuṇḍas* is found in front of the Kelaṇi Temple at Colombo. Here, a fountain has been set in the centre of a shallow multi-stepped concentric floor ornament imitating earlier diagrammatic designs of *kuṇḍas*.[57] Although the construction of *kuṇḍas* for the provision of water for the people was superseded by cheaper and quicker methods as water began to be transported over longer distances, the spectacular visual effect and the typical aesthetic achieved by their characteristic pyramidal step formations and multi-stepped and concentric nature have found some echoes in modern South Asian architecture.

[57] For more examples of modern *kuṇḍas* and a more detailed discussion of modern water architecture in general, see Chapter Eight, 'Conclusion'.